

Activity__Evaluate simple linear regression

June 29, 2025

1 Activity: Evaluate simple linear regression

1.1 Introduction

In this activity, you will use simple linear regression to explore the relationship between two continuous variables. To accomplish this, you will perform a complete simple linear regression analysis, which includes creating and fitting a model, checking model assumptions, analyzing model performance, interpreting model coefficients, and communicating results to stakeholders.

For this activity, you are part of an analytics team that provides insights about marketing and sales. You have been assigned to a project that focuses on the use of influencer marketing, and you would like to explore the relationship between marketing promotional budgets and sales. The dataset provided includes information about marketing campaigns across TV, radio, and social media, as well as how much revenue in sales was generated from these campaigns. Based on this information, leaders in your company will make decisions about where to focus future marketing efforts, so it is critical to have a clear understanding of the relationship between the different types of marketing and the revenue they generate.

This activity will develop your knowledge of linear regression and your skills evaluating regression results which will help prepare you for modeling to provide business recommendations in the future.

1.2 Step 1: Imports

1.2.1 Import packages

Import relevant Python libraries and packages. In this activity, you will need to use `pandas`, `pyplot` from `matplotlib`, and `seaborn`.

```
[3]: # Import pandas, pyplot from matplotlib, and seaborn.  
  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns
```

1.2.2 Import the statsmodel module and the ols function

Import the `statsmodels.api` Python module using its common abbreviation, `sm`, along with the `ols()` function from `statsmodels.formula.api`. To complete this, you will need to write the imports as well.

```
[4]: # Import the statsmodel module.

# Import the ols function from statsmodels.

from statsmodels.formula.api import ols
import statsmodels.api as sm
```

1.2.3 Load the dataset

Pandas was used to load the provided dataset `marketing_and_sales_data_evaluate_lr.csv` as `data`, now display the first five rows. This is a fictional dataset that was created for educational purposes. The variables in the dataset have been kept as is to suit the objectives of this activity. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the `.csv` file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[5]: # RUN THIS CELL TO IMPORT YOUR DATA.

data = pd.read_csv('marketing_and_sales_data_evaluate_lr.csv')

# Display the first five rows.

data.head(5)
```

```
[5]:      TV      Radio  Social_Media      Sales
0  16.0    6.566231    2.907983    54.732757
1  13.0    9.237765    2.409567    46.677897
2  41.0   15.886446    2.913410   150.177829
3  83.0   30.020028    6.922304   298.246340
4  15.0    8.437408    1.405998    56.594181
```

1.3 Step 2: Data exploration

1.3.1 Familiarize yourself with the data's features

Start with an exploratory data analysis to familiarize yourself with the data and prepare it for modeling.

The features in the data are: * TV promotion budget (in millions of dollars) * Social media promotion budget (in millions of dollars) * Radio promotion budget (in millions of dollars) * Sales

(in millions of dollars)

Each row corresponds to an independent marketing promotion where the business invests in **TV**, **Social_Media**, and **Radio** promotions to increase **Sales**.

The business would like to determine which feature most strongly predicts **Sales** so they have a better understanding of what promotions they should invest in in the future. To accomplish this, you'll construct a simple linear regression model that predicts sales using a single independent variable.

Question: What are some reasons for conducting an EDA before constructing a simple linear regression model?

Reasons for conducting an EDA includes:

- To understand which variables are present in the dataset.
- To understand the distribution of features such as minimum, mean and maximum values
- Plotting the relationship between the dependent and independent variables to visualize which feature is the best choice for x
- Identifying issues with the data such as incorrect values (i.e typos) and missing values

1.3.2 Explore the data size

Calculate the number of rows and columns in the data.

```
[6]: # Display the shape of the data as a tuple (rows, columns).  
  
data.shape
```

```
[6]: (4572, 4)
```

Hint 1

There is an attribute of a pandas DataFrame that returns the dimension of the DataFrame.

Hint 2

The **shape** attribute of a DataFrame returns a tuple with the array dimensions.

Hint 3

Use **data.shape**, which returns a tuple with the number of rows and columns.

1.3.3 Explore the independent variables

There are three continuous independent variables: **TV**, **Radio**, and **Social_Media**. To understand how heavily the business invests in each promotion type, use **describe()** to generate descriptive statistics for these three variables.

```
[7]: # Generate descriptive statistics about TV, Radio, and Social_Media.  
  
data.describe()
```

```
[7]:
```

	TV	Radio	Social_Media	Sales
count	4562.000000	4568.000000	4566.000000	4566.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.125054	9.676958	2.212670	93.133092
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.525957	1.527849	112.322882
50%	53.000000	17.859513	3.055565	189.231172
75%	77.000000	25.649730	4.807558	272.507922
max	100.000000	48.871161	13.981662	364.079751

Hint 1

Subset **data** to only include the columns of interest.

Hint 2

Select the columns of interest using `data[['TV', 'Radio', 'Social_Media']]`.

Hint 3

Apply `describe()` to the data subset.

1.3.4 Explore the dependent variable

Before fitting the model, ensure the **Sales** for each promotion (i.e., row) is present. If the **Sales** in a row is missing, that row isn't of much value to the simple linear regression model.

Display the percentage of missing values in the **Sales** column in the DataFrame **data**.

```
[8]: # Calculate the average missing rate in the sales column.

missing_sales = data.Sales.isna().mean()

# Convert the missing_sales from a decimal to a percentage and round to 2
↳ decimal place.

missing_sales = round(missing_sales*100,2)

# Display the results (missing_sales must be converted to a string to be
↳ concatenated in the print statement).

print("Percentage of promotions missing sales: " + str(missing_sales) + "%")
```

Percentage of promotions missing sales: 0.13%

Question: What do you observe about the percentage of missing values in the **Sales** column?

The preceding output showw that 0.13% of rows are missing the **Sales** value.

1.3.5 Remove the missing data

Remove all rows in the data from which **Sales** is missing.

```
[9]: # Subset the data to include rows where Sales is present.  
  
data = data.dropna(subset = ['Sales'], axis = 0)
```

Hint 1

Refer to [the content about removing missing values from a DataFrame](#).

Hint 2

The `dropna()` function may be helpful.

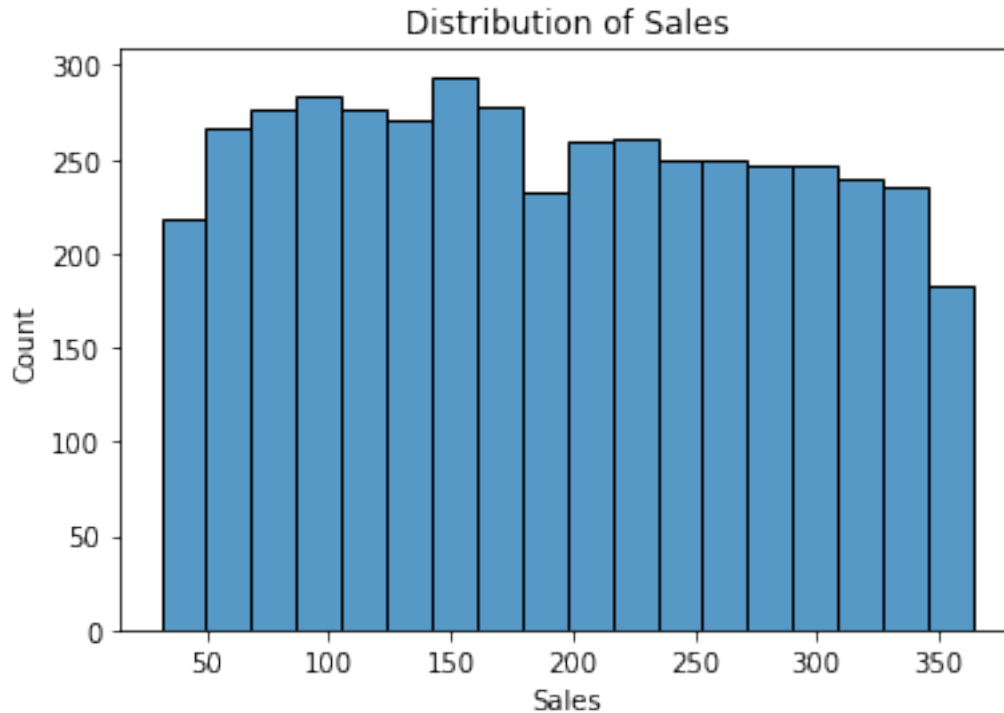
Hint 3

Apply `dropna()` to `data` and use the `subset` and `axis` arguments to drop rows where **Sales** is missing.

1.3.6 Visualize the sales distribution

Create a histogram to visualize the distribution of **Sales**.

```
[10]: # Create a histogram of the Sales.  
  
fig = sns.histplot(data["Sales"])  
  
# Add a title  
  
fig.set_title("Distribution of Sales");
```



Hint 1

Use the function in the `seaborn` library that allows you to create a histogram.

Hint 2

Call the `histplot()` function from the `seaborn` library and pass in the `Sales` column as the argument.

Hint 3

To get a specific column from a `DataFrame`, use a pair of single square brackets and place the name of the column, as a string, in the brackets. Be sure that the spelling, including case, matches the data exactly.

Question: What do you observe about the distribution of `Sales` from the preceding histogram?

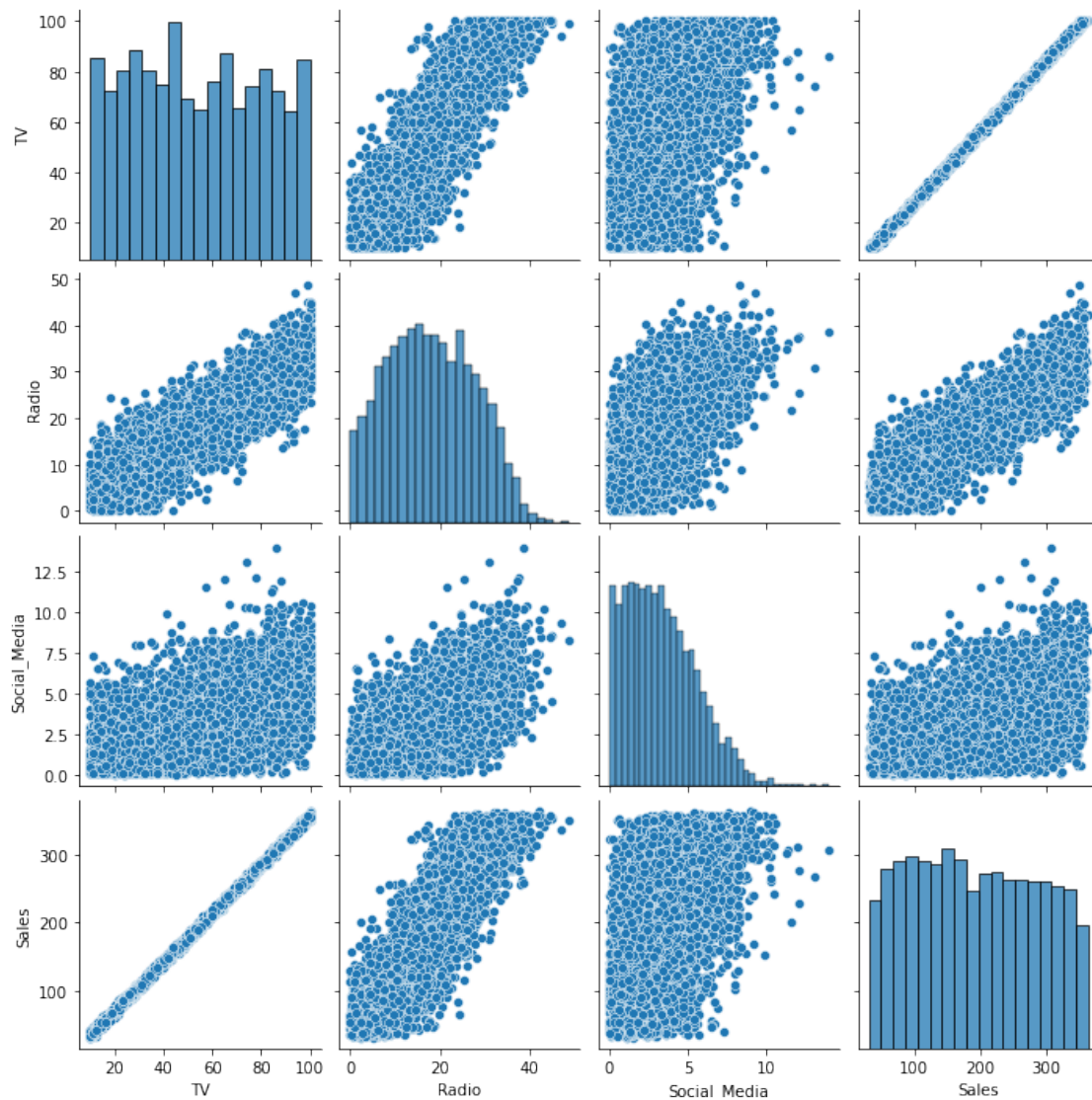
Generally, `Sales` are equally distributed between 25 and 350 million.

1.4 Step 3: Model building

Create a pairplot to visualize the relationships between pairs of variables in the data. You will use this to visually determine which variable has the strongest linear relationship with `Sales`. This will help you select the X variable for the simple linear regression.

```
[15]: # Create a pairplot of the data.
```

```
sns.pairplot(data);
```



Hint 1

Refer to [the video](#) where creating a pairplot is demonstrated.

Hint 2

Use the function in the **seaborn** library that allows you to create a pairplot that shows the relationships between variables in the data.

Hint 3

Use the `pairplot()` function from the **seaborn** library and pass in the entire DataFrame.

Question: Which variable did you select for X? Why?

TV clearly has the strongest linear relationship with sales. You could draw a straight line through the scatterplot TV and Sales that confidently estimates the Sales using TV. Radio and Sales appear to have a linear relationship, but there is a larger linear variance than between TV and Sales.

1.4.1 Build and fit the model

Replace the comment with the correct code. Use the variable you chose for X for building the model.

```
[22]: # Define the OLS formula.

ols_formula = 'Sales ~ TV'

# Create an OLS model.

OLS =ols(formula= ols_formula, data=data)

# Fit the model.

model = OLS.fit()

# Save the results summary.

model_result = model.summary()

# Display the model results.

model_result
```

```
[22]: <class 'statsmodels.iolib.summary.Summary'>
"""
                                OLS Regression Results
=====
Dep. Variable:                  Sales    R-squared:                0.999
Model:                            OLS    Adj. R-squared:            0.999
Method:                 Least Squares    F-statistic:                4.527e+06
Date:                Fri, 27 Jun 2025    Prob (F-statistic):          0.00
Time:                  03:31:03    Log-Likelihood:             -11393.
No. Observations:                4556    AIC:                        2.279e+04
Df Residuals:                    4554    BIC:                        2.280e+04
Df Model:                            1
Covariance Type:                nonrobust
=====
                                coef    std err          t      P>|t|    [0.025    0.975]
-----

```


Intercept	-0.1263	0.101	-1.257	0.209	-0.323	0.071
TV	3.5614	0.002	2127.776	0.000	3.558	3.565

```
=====
```

Omnibus:	0.051	Durbin-Watson:	2.002
Prob(Omnibus):	0.975	Jarque-Bera (JB):	0.030
Skew:	0.001	Prob(JB):	0.985
Kurtosis:	3.012	Cond. No.	138.

```
=====
```

Warnings:

```
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
"""
```

Hint 1

Refer to [the video where an OLS model is defined and fit](#).

Hint 2

Use the `ols()` function imported earlier— which creates a model from a formula and DataFrame—to create an OLS model.

Hint 3

Replace the `X` in `'Sales ~ X'` with the independent feature you determined has the strongest linear relationship with `Sales`. Be sure the string name for `X` exactly matches the column's name in `data`.

Hint 4

Obtain the model results summary using `model.summary()` and save it. Be sure to fit the model before saving the results summary.

1.4.2 Check model assumptions

To justify using simple linear regression, check that the four linear regression assumptions are not violated. These assumptions are:

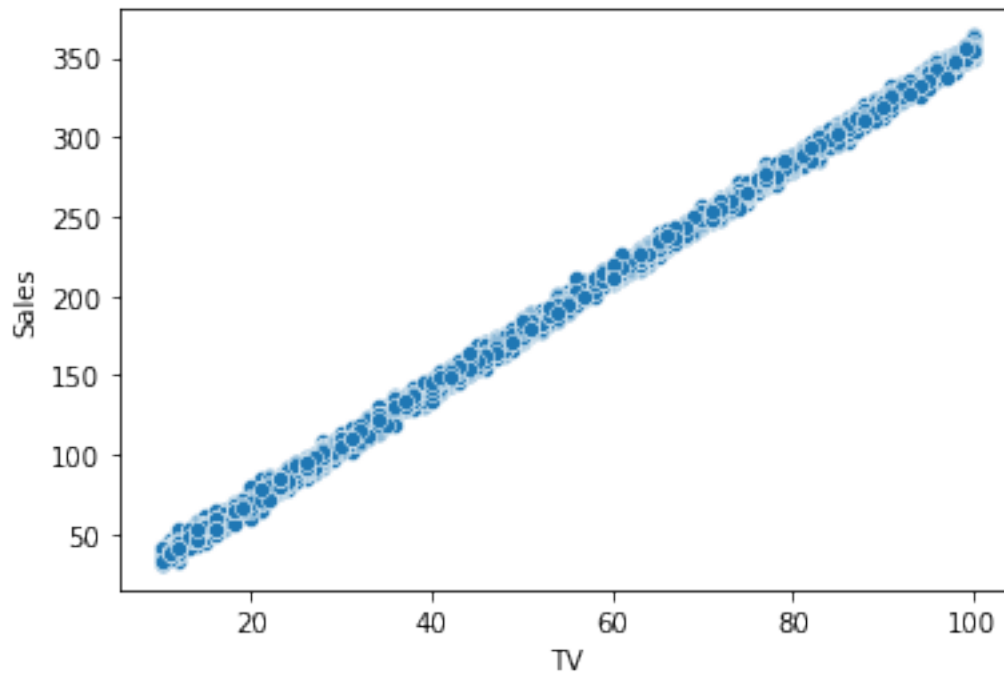
- Linearity
- Independent Observations
- Normality
- Homoscedasticity

1.4.3 Model assumption: Linearity

The linearity assumption requires a linear relationship between the independent and dependent variables. Check this assumption by creating a scatterplot comparing the independent variable with the dependent variable.

Create a scatterplot comparing the `X` variable you selected with the dependent variable.

```
[17]: # Create a scatterplot comparing X and Sales (Y).  
  
sns.scatterplot(x =data['TV'], y= data['Sales']);
```



Hint 1

Use the function in the **seaborn** library that allows you to create a scatterplot to display the values for two variables.

Hint 2

Use the `scatterplot()` function in **seaborn**.

Hint 3

Pass the X and Y variables you chose for your simple linear regression as the arguments for **x** and **y**, respectively, in the `scatterplot()` function.

QUESTION: Is the linearity assumption met?

There is a clear linear relationship between TV and Sales, meeting the linearity assumption.

1.4.4 Model assumption: Independence

The **independent observation assumption** states that each observation in the dataset is independent. As each marketing promotion (i.e., row) is independent from one another, the independence assumption is not violated.

1.4.5 Model assumption: Normality

The normality assumption states that the errors are normally distributed.

Create two plots to check this assumption:

- **Plot 1:** Histogram of the residuals
- **Plot 2:** Q-Q plot of the residuals

```
[18]: # Calculate the residuals.

residuals = model.resid

# Create a 1x2 plot figures.

fig, axes = plt.subplots(1, 2, figsize = (8,4))

# Create a histogram with the residuals.

sns.histplot(residuals, ax=axes[0])

# Set the x label of the residual plot.

axes[0].set_xlabel("Residual Value")

# Set the title of the residual plot.

axes[0].set_ylabel("Histogram of Residuals")

# Create a Q-Q plot of the residuals.

sm.qqplot(residuals, line='s', ax = axes[1])

# Set the title of the Q-Q plot.

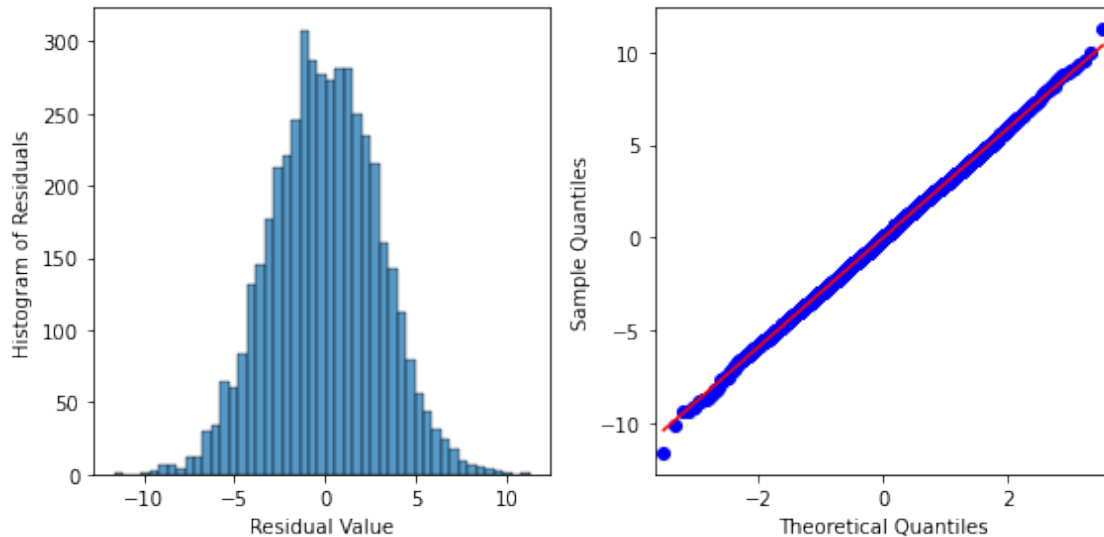
axes[1].set_label('Normal Q-Q plot')

# Use matplotlib's tight_layout() function to add space between plots for a
↳ cleaner appearance.

plt.tight_layout()

# Show the plot.

plt.show()
```



Hint 1

Access the residuals from the fit model object.

Hint 2

Use `model.resid` to get the residuals from the fit model.

Hint 3

For the histogram, pass the residuals as the first argument in the `seaborn histplot()` function.

For the Q-Q plot, pass the residuals as the first argument in the `statsmodels qqplot()` function.

Question: Is the normality assumption met?

The histogram of the residuals are approximately normally distributed, which supports that the normality assumption is met for this model. The residuals in the Q-Q plot forms a straight line, further supporting that the normality notion is met.

1.4.6 Model assumption: Homoscedasticity

The **homoscedasticity (constant variance) assumption** is that the residuals have a constant variance for all values of X .

Check that this assumption is not violated by creating a scatterplot with the fitted values and residuals. Add a line at $y = 0$ to visualize the variance of residuals above and below $y = 0$.

```
[19]: # Create a scatterplot with the fitted values from the model and the residuals.

fig = sns.scatterplot(x = model.fittedvalues, y = model.resid)

# Set the x-axis label.
```

```

fig.set_xlabel("Fitted Values")

# Set the y-axis label.

fig.set_ylabel("Residuals")

# Set the title.

fig.set_title("Fitted Vlaues vs. Residuals")

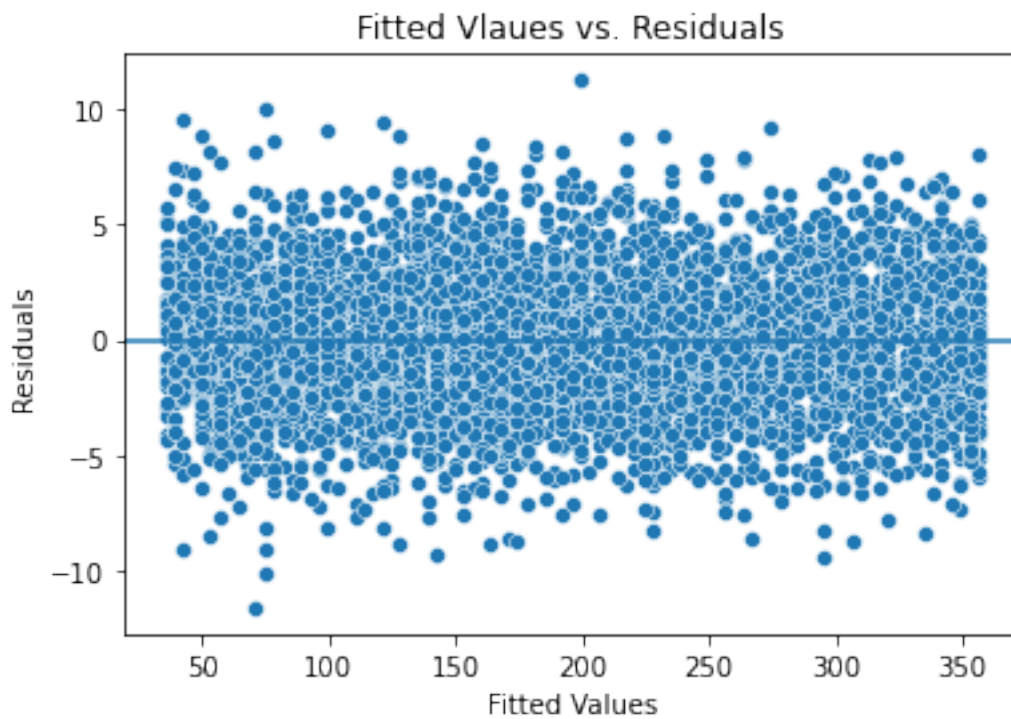
# Add a line at y = 0 to visualize the variance of residuals above and below 0.

fig.axhline(0)

# Show the plot.

plt.show()

```



Hint 1

Access the fitted values from the `model` object fit earlier.

Hint 2

Use `model.fittedvalues` to get the fitted values from the fit model.

Hint 3

Call the `scatterplot()` function from the `seaborn` library and pass in the fitted values and residuals.

Add a line to the figure using the `axline()` function.

QUESTION: Is the homoscedasticity assumption met?

The variance of residuals is consistent across all X . Thus the assumption of homoscedasticity is met.

1.5 Step 4: Results and evaluation

1.5.1 Display the OLS regression results

If the linearity assumptions are met, you can interpret the model results accurately.

Display the OLS regression results from the fitted model object, which includes information about the dataset, model fit, and coefficients.

```
[23]: # Display the model_results defined previously.
```

```
model_result
```

```
[23]: <class 'statsmodels.iolib.summary.Summary'>
```

```
"""
                                OLS Regression Results
=====
Dep. Variable:                  Sales    R-squared:                        0.999
Model:                            OLS    Adj. R-squared:                   0.999
Method:                 Least Squares    F-statistic:                   4.527e+06
Date:                Fri, 27 Jun 2025    Prob (F-statistic):              0.00
Time:                  03:31:03    Log-Likelihood:                -11393.
No. Observations:                  4556    AIC:                          2.279e+04
Df Residuals:                      4554    BIC:                          2.280e+04
Df Model:                            1
Covariance Type:                  nonrobust
=====
               coef    std err          t      P>|t|      [0.025    0.975]
-----
Intercept    -0.1263     0.101     -1.257     0.209    -0.323     0.071
TV             3.5614     0.002   2127.776     0.000     3.558     3.565
=====
Omnibus:                 0.051    Durbin-Watson:                   2.002
Prob(Omnibus):            0.975    Jarque-Bera (JB):                 0.030
Skew:                     0.001    Prob(JB):                         0.985
Kurtosis:                 3.012    Cond. No.                         138.
=====
```

=====

Warnings:

```
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
"""
```

Question: The R-squared on the preceding output measures the proportion of variation in the dependent variable (Y) explained by the independent variable (X). What is your interpretation of the model's R-squared?

Using TV as X results in a simple linear regression model with $R^2=0.999$.

In other words, TV explains 99.9% of the variation in Sales.

T-squared value will depend on the variable selected for x.

1.5.2 Interpret the model results

With the model fit evaluated, assess the coefficient estimates and the uncertainty of these estimates.

Question: Based on the preceding model results, what do you observe about the coefficients?

When TV is used as the independent variable X, the coefficient for the **Intercept** is -0.1263 and the coefficient for TV is 3.5614.

Question: How would you write the relationship between X and Sales in the form of a linear equation?

When TV is used as the independent variable X, the linear equation is:

$$Y = \text{Intercept} + \text{Slope} * X$$

$$\text{Sales (in millions)} = \text{Intercept} + \text{Slope} * \text{TV (in millions)}$$

$$\text{Sales (in millions)} = -0.1263 + 3.5614 * \text{TV (in millions)}$$

Question: Why is it important to interpret the beta coefficients?

Beta coefficients allow you to estimate the magnitude and direction (positive or negative) of the effect of each independent variable on the dependent variable. The coefficient estimates can be converted to explainable insights, such as the connection between an increase in TV promotional budgets and sales mentioned previously.

1.5.3 Measure the uncertainty of the coefficient estimates

Model coefficients are estimated. This means there is an amount of uncertainty in the estimate. A p-value and 95% confidence interval are provided with each coefficient to quantify the uncertainty for that coefficient estimate.

Display the model results again.

```
[25]: # Display the model_results defined previously.
```

```
model_result
```

```
[25]: <class 'statsmodels.iolib.summary.Summary'>
```

```
"""
                                OLS Regression Results
=====
Dep. Variable:                  Sales    R-squared:                        0.999
Model:                            OLS    Adj. R-squared:                    0.999
Method:                 Least Squares    F-statistic:                  4.527e+06
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Df Residuals:                    4554    BIC:                           2.280e+04
Df Model:                            1
Covariance Type:                nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    -0.1263      0.101     -1.257      0.209     -0.323      0.071
TV             3.5614      0.002    2127.776      0.000      3.558      3.565
=====
Omnibus:                        0.051    Durbin-Watson:                   2.002
Prob(Omnibus):                   0.975    Jarque-Bera (JB):                 0.030
Skew:                            0.001    Prob(JB):                        0.985
Kurtosis:                       3.012    Cond. No.                        138.
=====

Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
"""
```

Question: Based on this model, what is your interpretation of the p-value and confidence interval for the coefficient estimate of X?

When TV is used as the independent variable, it has a p-value of 0.000 and a 95% confidence interval of [3.558, 3.565]. This means there is a 95% chance the interval [3.558, 3.565] contains the true parameter value of the slope. These results indicate little uncertainty in the estimation of the slope of X. Therefore, the business can be confident in the impact TV has on Sales.

Question: Based on this model, what are you interested in exploring?

- Providing the business with the estimated sales given different TV promotional budgets
- Using both TV and Radio as independent variable
- Adding plots to help convey the results, such as using the `seaborn regplot()` to plot the data with a best fit regression line

Question: What recommendations would you make to the leadership at your organization?

Of the three available promotion types (TV, radio, and social media), TV has the strongest positive linear relationship with sales. According to the model, an increase of one million dollars for the TV promotional budget will result in an estimated 3.5614 million dollars more in sales. This is a very confident estimate, as the p-value for this coefficient estimate is small. Thus, the business should prioritize increasing the TV promotional budget over the radio and social media promotional budget to increase sales.

1.6 Considerations

What are some key takeaways that you learned from this lab?

- Exploratory data analytics can be used to find a suitable X variable for a simple linear regression model.
- It's important to check that simple linear regression assumptions are met before interpreting model results.
- R-squared is a useful measure of prediction error.
- It's important to provide measures of uncertainty (e.g., p-values, confidence intervals) with any coefficient estimates.

What findings would you share with others?

- Sales is relatively equally distributed between \$25 and \$350 million across all promotions.
- In data visualization, TV has the strongest linear relationship with sales. Radio and sales appear to have a moderate linear relationship, but there is larger variance than between TV and sales. Social media and sales appear to have a weak linear relationship.
- When TV is used as the independent variable, the R-squared for the simple linear regression model estimating sales is 0.999.
- When TV is used as the independent variable, the coefficient for the intercept is -0.1263 and the coefficient for TV is 3.5614.
- When TV is used as the independent variable, the slope of the coefficient for TV has a p-value of 0.000 and a 95% confidence interval of [3.558, 3.565].

How would you frame your findings to stakeholders?

- Among TV, social media, and radio, TV had the strongest positive linear relationship with sales.
- The linear regression model estimates that 99.9% of the variation in sales is explained by the TV promotion budget. In other words, nearly all of the variation in sales can be explained by the TV promotion budget alone, making TV an excellent predictor of sales.
- According to the model, when TV is used as independent variable X, an increase of one million dollars for the TV promotional budget would result in an estimated 3.5614 million more dollars in sales.
- The interval (3.558 million, 3.565 million) has a 95% probability of containing the true estimate of the increase in sales for one million dollar increase in the TV promotional budget. Therefore, the estimate provided in the previous bullet is very confident.

References Saragih, H.S. (2020). *Dummy Marketing and Sales Data*.

Dale, D., Droettboom, M., Firing, E., Hunter, J. (n.d.). [Matplotlib.Pyplot.Axline — Matplotlib 3.5.0 Documentation](#).

Congratulations! You’ve completed this lab. However, you may not notice a green check mark next to this item on Coursera’s platform. Please continue your progress regardless of the check mark. Just click on the “save” icon at the top of this notebook to ensure your work has been logged.